✅ SUPERVISED LEARNING — Evaluation Metrics

🔹 Classification Metrics

📊 Core Classification Metrics

Metric	Used For	What It Checks (Plain English)	When To Use	Common Mistake
Accuracy	All classifiers	Overall correctness	Balanced classes	Misleading with imbalanced data
Precision	Binary / multi	“When I predict positive, am I right?”	False positives are costly	Ignoring recall
Recall (Sensitivity)	Binary / multi	“Did I catch all real positives?”	Missing positives is costly	Ignoring precision
F1-score	Binary / multi	Balance of precision & recall	Imbalanced data	Using accuracy instead
Confusion Matrix	All classifiers	Breakdown of correct vs wrong predictions	Model diagnosis	Looking only at accuracy

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🧠 Advanced Classification Metrics

Metric	Used For	What It Checks	When To Use
ROC-AUC	Probabilistic models	Ranking quality across thresholds	Binary classifiers
Log Loss	Probabilistic models	Confidence of predictions	Probability outputs
PR-AUC	Imbalanced datasets	Precision–Recall tradeoff	Rare positive cases
Balanced Accuracy	Imbalanced data	Mean recall per class	Skewed classes

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🏭 Algorithm → Metric Mapping (Classifier)

Algorithm	Recommended Metrics
Logistic Regression	Accuracy, Precision, Recall, F1, ROC-AUC
Decision Tree	Accuracy, F1, Confusion Matrix
Random Forest	F1, ROC-AUC, PR-AUC
SVM	Accuracy, F1, ROC-AUC (if probability enabled)
Naive Bayes	Precision, Recall

📈 REGRESSION — Evaluation Metrics

Metric	What It Checks	Interpretation	When To Use
MSE	Average squared error	Penalizes big errors heavily	Training optimization
RMSE	Error in original units	Easy to interpret	Reporting results
MAE	Average absolute error	Robust to outliers	Noisy data
R^2	Variance explained	Model goodness	Model comparison
Adjusted R^2	Penalizes extra features	Overfitting control	Linear regression

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🔍 UNSUPERVISED LEARNING — Evaluation Metrics

🔹 Clustering Metrics (NO labels)

📊 Internal Evaluation Metrics

Metric	Algorithm	What It Checks	Interpretation
SSE (Inertia)	K-Means	Cluster compactness	Lower = tighter clusters
Silhouette Score ⭐	Most clustering	Cohesion vs separation	-1 to +1
Davies–Bouldin Index	K-Means, others	Cluster overlap	Lower = better
Calinski–Harabasz	Clustering	Between vs within variance	Higher = better

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🔹 Clustering Metrics (WITH labels, not for training)

Metric	What It Checks	Why It's Special
Adjusted Rand Index (ARI) ⭐	Label agreement	Label-order invariant
Normalized Mutual Info (NMI)	Shared information	Scale-independent
Homogeneity	Each cluster = one class	Pure clusters
Completeness	Each class = one cluster	No split classes
V-measure	Harmonic of above	Balanced clustering

🔹 Algorithm → Metric Mapping (Unsupervised)

Algorithm	Metrics You Should Use
K-Means	SSE, Silhouette, Davies–Bouldin
Hierarchical	Silhouette, Dendrogram
DBSCAN	Silhouette, Noise Ratio
Mean Shift	Silhouette
PCA	Explained Variance Ratio

🧠 DIMENSIONALITY REDUCTION (Special Case)

Metric	What It Checks	Used With
Explained Variance	Information retained	PCA
Cumulative Variance	Compression quality	PCA
Reconstruction Error	Information loss	Autoencoders

🚨 Common Metric Mistakes (VERY important)

❌ Using accuracy for clustering
❌ Using only SSE to pick K

❌ Ignoring recall in medical/fraud models
❌ Using R^2 alone for regression
❌ Evaluating on training data

🧠 **Golden Rule (memorize this)**

Prediction → Accuracy-based metrics
Distance/structure → Geometry-based metrics
Probability → AUC / Log loss
No labels → Silhouette-style metrics